

Lab 3: Architectural Justification

PES1UG24CS125 | Charan M | Section B | Problem Statement #15: Pharmacy Expiry & Re-order Dispatch Engine

Architecture Selection: We chose **Layered Architecture** for the Pharmacy Expiry & Re-order Dispatch Engine.

Architectural Choice

The system is organised into three layers. The Presentation layer holds the Pharmacy Management UI. The Business layer holds the Dispensing & FEFO Engine, Inventory Management, Re-order / Purchase Order Service, Expiry Alert Service and Audit Trail Service. The Data layer holds the Pharmacy Database. The Inventory Supplier is an external system reached only through the Supplier Interface.

Reason 1: The core workflow is a top-down chain

Dispensing medicine (UC-01) flows from the clerk to the UI, then to the Dispensing & FEFO Engine (UC-02), then to Inventory Management and finally the database. Each step depends only on the one below it (Interfaces 1, 2 and 3), which maps directly onto the layers. The FEFO and override rules (FR-001, FR-004) live in a single business component, so the UI cannot bypass them.

Reason 2: The alternatives do not fit this scale or these requirements

This is a pharmacy system used by a small number of clerks, and no requirement calls for independently scaled services. Microservices would add network latency and distributed-consistency risk to stock deduction, where batch quantities must stay exact, and would add operational complexity with no benefit. Client-Server only describes where code runs, not how responsibilities are separated, so it would give a much weaker component design. Layered gives clear separation of concerns and makes each component easy to test and replace behind its interface.

Security Advantage

The UI never touches the database. All data access passes through the Business layer and one database interface (Interface 3), so authorisation checks are enforced in one place. Every dispensing transaction is sent to the Audit Trail Service through Interface 5, which supports the tamper-evident, 3-year audit trail required by NFR-002. Purchase orders leave the system through a single controlled channel (Interface 6, HTTPS), which makes the automated dispatch in NFR-001 easier to secure and monitor.

Performance Benefit

The layers are coarse-grained and deployed as one application, so a checkout needs only a few local calls and no network hops between internal services. FEFO ranking is a single query for a medicine's unexpired batches ordered by expiry date, which keeps checkout fast. Purchase order dispatch runs only when stock drops below the safety threshold and is a small operation, so the extra layer overhead does not slow dispensing.

Trade-off Acknowledged

Layered architecture is harder to scale piece by piece and adds some call overhead between layers. For a single-pharmacy workload this is acceptable, and it is outweighed by the simplicity, traceability and data integrity the design provides.